

Clairaut's Relation for Geodesics on Surfaces of Revolution

Faris Doughan

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Abstract

Surfaces of revolution form one of the simplest and most useful classes of surfaces in differential geometry. Their rotational symmetry leads to a conserved quantity along geodesics, known as Clairaut's relation. In this paper we introduce surfaces of revolution in $\mathbb{R}^3$, set up the geodesic equations in adapted coordinates, and prove Clairaut's relation.

1 Introduction

Geodesics are curves that locally minimize distance and are the natural generalization of straight lines to curved spaces. On a Riemannian surface they describe the motion of a particle that moves with constant speed and no external forces. In general, the geodesic equations are second order nonlinear differential equations and can be difficult to solve explicitly.

On some surfaces there is additional symmetry that simplifies the problem. An important example is a surface of revolution, obtained by rotating a plane curve around an axis. The symmetry under rotation around the axis leads to a conserved quantity along geodesics. This conserved quantity is Clairaut's relation.

Informally, Clairaut's relation says that on a surface of revolution, if r denotes the distance from a point to the axis of revolution and α is the angle between the geodesic and a meridian (a curve going straight toward the axis), then the product $r \sin \alpha$ is constant along the geodesic. As the geodesic moves toward the axis and r decreases, the angle α must increase so that $\sin \alpha$ compensates.

2 Surfaces of revolution

We start by fixing notation and recalling the basic formulas for surfaces of revolution.

Definition 2.1. Let $\gamma: I \rightarrow \mathbb{R}^2$ be a smooth plane curve of the form

$$\gamma(u) = (f(u), g(u)), \quad u \in I \subset \mathbb{R},$$

with $f(u) > 0$ for all $u \in I$. The surface obtained by rotating γ around the z -axis in $\mathbb{R}^3$ is the surface of revolution parametrized by

$$X(u, v) = (f(u) \cos v, f(u) \sin v, g(u)), \quad (u, v) \in I \times (0, 2\pi).$$

Here u measures position along the generating curve and v is the rotation angle.

We think of u as the “meridian” coordinate and v as the “parallel” coordinate. For fixed u , the curve $v \mapsto X(u, v)$ is a circle of radius $f(u)$ around the z -axis. For fixed v , the curve $u \mapsto X(u, v)$ is a copy of the generating curve lying in a meridian plane.

First fundamental form

To study geodesics we need the induced Riemannian metric on the surface. This is given by the first fundamental form in the coordinates (u, v) .

We compute the tangent vectors

$$X_u = \frac{\partial X}{\partial u} = (f'(u) \cos v, f'(u) \sin v, g'(u)),$$

$$X_v = \frac{\partial X}{\partial v} = (-f(u) \sin v, f(u) \cos v, 0).$$

The coefficients of the first fundamental form are

$$E(u) = \langle X_u, X_u \rangle = f'(u)^2 + g'(u)^2,$$

$$F(u) = \langle X_u, X_v \rangle = 0,$$

$$G(u) = \langle X_v, X_v \rangle = f(u)^2,$$

Remark 2.2. The cross term F vanishes for every surface of revolution in these coordinates. This means the coordinate lines $u = \text{const}$ and $v = \text{const}$ are orthogonal. The coefficient $G(u) = f(u)^2$ is the square of the distance $f(u)$ from the point to the axis of revolution. The function $E(u)$ encodes the speed of the generating curve.

The induced Riemannian metric on the surface is therefore

$$ds^2 = E(u) du^2 + G(u) dv^2 = (f'(u)^2 + g'(u)^2) du^2 + f(u)^2 dv^2.$$

Arc length of curves

Let $c: [a, b] \rightarrow I \times (0, 2\pi)$ be a smooth curve on the parameter domain, written as

$$c(t) = (u(t), v(t)).$$

The corresponding curve on the surface is

$$\alpha(t) = X(u(t), v(t)).$$

The velocity vector of α is

$$\alpha'(t) = u'(t)X_u(u(t), v(t)) + v'(t)X_v(u(t), v(t)),$$

so its squared speed is

$$\begin{aligned} \|\alpha'(t)\|^2 &= E(u(t)) u'(t)^2 + 2F(u(t)) u'(t)v'(t) + G(u(t)) v'(t)^2 \\ &= E(u(t)) u'(t)^2 + G(u(t)) v'(t)^2, \end{aligned}$$

since $F = 0$. The length of α is

$$L(\alpha) = \int_a^b \sqrt{E(u(t)) u'(t)^2 + G(u(t)) v'(t)^2} dt.$$

Geodesics are curves that locally minimize this length functional and can be described as critical points of L .

3 Geodesics and Clairaut's relation

To find geodesics it is convenient to use the language of calculus of variations. We treat L as the length functional for curves in the (u, v) plane with the metric $E du^2 + G dv^2$ and derive the Euler–Lagrange equations. The rotational symmetry will give a conserved quantity.

Geodesic equations from the energy functional

Instead of working directly with L , it is standard to use the energy functional

$$\mathcal{E}(\alpha) = \frac{1}{2} \int_a^b (E(u) u'(t)^2 + G(u) v'(t)^2) dt.$$

Critical points of $\mathcal{E}$ with fixed endpoints are the same as critical points of L when curves are parametrized by arc length. The advantage is that $\mathcal{E}$ has a simpler integrand.

Define the Lagrangian

$$L(u, v, u', v') = \frac{1}{2} (E(u) u'^2 + G(u) v'^2).$$

The Euler–Lagrange equations are

$$\frac{d}{dt} \left(\frac{\partial L}{\partial u'} \right) - \frac{\partial L}{\partial u} = 0, \quad \frac{d}{dt} \left(\frac{\partial L}{\partial v'} \right) - \frac{\partial L}{\partial v} = 0.$$

We compute

$$\frac{\partial L}{\partial u'} = E(u) u', \quad \frac{\partial L}{\partial v'} = G(u) v',$$

and

$$\frac{\partial L}{\partial u} = \frac{1}{2} E'(u) u'^2 + \frac{1}{2} G'(u) v'^2, \quad \frac{\partial L}{\partial v} = 0,$$

since E and G depend only on u .

The equation corresponding to v is therefore

$$\frac{d}{dt} (G(u) v'(t)) = 0.$$

This implies that the quantity

$$G(u(t)) v'(t) = f(u(t))^2 v'(t)$$

is constant along the curve. This is the key conserved quantity.

Proposition 3.1. *Let $\alpha(t) = X(u(t), v(t))$ be a geodesic on a surface of revolution parametrized as above. Then there exists a constant $c \in \mathbb{R}$ such that*

$$f(u(t))^2 v'(t) = c$$

for all t .

Proof. As shown above, the geodesic equations for the energy functional imply that

$$\frac{d}{dt} (G(u) v'(t)) = 0.$$

Hence $G(u)v'$ is constant along the geodesic. Since $G(u) = f(u)^2$, we obtain the desired relation. $\square$

Clairaut's relation

To express the conserved quantity in a more geometric way we introduce the angle that the geodesic makes with the meridians.

Let α be parametrized by arc length, so that $\|\alpha'(s)\| \equiv 1$. Then

$$1 = \|\alpha'(s)\|^2 = E(u(s)) u'(s)^2 + G(u(s)) v'(s)^2.$$

At a point $\alpha(s)$ the meridian through $\alpha(s)$ is the curve $t \mapsto X(u(s) + t, v(s))$. Its tangent direction is $X_u(u(s), v(s))$. The unit vector in the meridian direction is

$$e_u(s) = \frac{X_u(u(s), v(s))}{\|X_u(u(s), v(s))\|} = \frac{X_u}{\sqrt{E(u(s))}}.$$

Definition 3.2. Let $\alpha(s)$ be a unit speed curve on the surface and $e_u(s)$ the unit meridian direction at $\alpha(s)$. The angle $\alpha(s)$ (we reuse the letter temporarily) between the geodesic and the meridian at $\alpha(s)$ is defined by

$$\cos \alpha(s) = \langle \alpha'(s), e_u(s) \rangle.$$

To avoid confusion with the curve itself we now denote the angle by $\theta(s)$.

Write $\alpha'(s) = u'(s)X_u + v'(s)X_v$ and use the inner product on the surface:

$$\langle X_u, X_u \rangle = E(u), \quad \langle X_v, X_v \rangle = G(u), \quad \langle X_u, X_v \rangle = 0.$$

Then

$$\langle \alpha'(s), e_u(s) \rangle = \left\langle u'(s)X_u + v'(s)X_v, \frac{X_u}{\sqrt{E(u)}} \right\rangle = u'(s)\sqrt{E(u(s))}.$$

Hence

$$\cos \theta(s) = u'(s)\sqrt{E(u(s))}.$$

Since α has unit speed, we also have

$$1 = E(u) u'^2 + G(u) v'^2.$$

Solving for $G(u)v'^2$ gives

$$G(u) v'^2 = 1 - E(u) u'^2 = 1 - \cos^2 \theta(s) = \sin^2 \theta(s).$$

We may take square roots and choose the sign so that

$$\sqrt{G(u)} |v'(s)| = |\sin \theta(s)|.$$

If we fix an orientation of the parameter v and the arc length parameter this becomes

$$\sqrt{G(u)} v'(s) = \sin \theta(s) \quad \text{or} \quad \sqrt{G(u)} v'(s) = -\sin \theta(s),$$

depending on the choice. For Clairaut's relation we care only about the absolute values.

From the conserved quantity we know that $G(u)v'(s) = c$ is constant. Taking absolute values and combining with the formula above yields

$$|G(u)v'(s)| = \sqrt{G(u)} |\sqrt{G(u)}v'(s)| = \sqrt{G(u)} |\sin \theta(s)| = \text{constant}.$$

Recall that $G(u) = f(u)^2$ and $f(u)$ is the distance from the point to the axis of revolution.

Theorem 3.3 (Clairaut’s relation). *Let S be a surface of revolution with parametrization*

$$X(u, v) = (f(u) \cos v, f(u) \sin v, g(u)), \quad f(u) > 0.$$

Let $\alpha(s)$ be a unit speed geodesic on S and let $r(s) = f(u(s))$ be the distance from $\alpha(s)$ to the axis of revolution. Let $\theta(s)$ be the angle between $\alpha'(s)$ and the meridian direction at $\alpha(s)$. Then

$$r(s) \sin \theta(s) = C$$

for some constant C independent of s .

Proof. As above we have $G(u) = f(u)^2$, so

$$\sqrt{G(u(s))} = f(u(s)) = r(s).$$

From the geodesic equation we know $G(u)v'(s)$ is constant, hence $|G(u)v'(s)|$ is constant. The unit speed condition gives

$$G(u)v'(s)^2 = \sin^2 \theta(s).$$

Taking square roots and multiplying by $\sqrt{G(u)}$ we obtain

$$|G(u)v'(s)| = \sqrt{G(u)} |\sin \theta(s)| = r(s) |\sin \theta(s)|.$$

The left side is constant in s , so $r(s) |\sin \theta(s)|$ is constant. Writing C for this constant and choosing a consistent sign convention gives the claimed relation. $\square$

Clairaut’s relation says that as a geodesic moves along a surface of revolution, the quantity “distance to the axis times sine of the entry angle” is preserved. It gives strong information about the qualitative shape of geodesics, even when we cannot solve the geodesic equations explicitly.

References

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